Stories of Your Life as Others: A Round-Trip Evaluation of LLM-Generated Life Stories Conditioned on Rich Psychometric Profiles

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Abstract

Language models are increasingly used to create simulated participants and personas for social-science research, but most personality evaluations ask only whether a model can answer questionnaires or follow short trait labels. We ask a stricter question: can measured psychometric profiles survive transformation into open-ended narrative? Using 290 participants from a social-interaction dataset with self-report psychometrics, we convert each participant's measured profile into controlled prose, generate Life Story Interview-style first-person answers, and ask independent models to recover the original scores from the generated text alone. Generated 24-turn life-story responses preserve substantial individual-difference signal: independent scorers recover HEXACO traits at $r = .761–.787$ and broader trust, psychopathy, social-anxiety, and social-value-orientation targets at lower but meaningful levels. Additional tests rule out several simple shortcuts, show signal in base/pretrained generators, and provide bounded human-reader and participant-language references. Generated narrative can preserve measurable psychometric structure, but the assay measures profile-to-language preservation rather than whether synthetic life stories replace human life-story interviews.

1 Introduction

Personality is one of the central cases where language and individual differences meet. Traits are robust predictors of life outcomes (Roberts et al., 2007; Ozer & Benet-Martínez, 2006), and trait-relevant information appears in word choice, emotional stance, interpersonal style, self-explanation, and the kinds of episodes people choose to narrate. This connection motivates a long tradition of personality prediction from text, from the lexical hypothesis (Saucier & Goldberg, 1996; Boyd & Pennebaker, 2017) to modern computational models trained on interviews, essays, social media, and self-narratives (Fast & Funder, 2008; Dai et al., 2022; Oltmanns et al., 2026; Speer et al., 2026; Wright et al., 2026).

Large language models make the problem newly bidirectional. They are not only used to infer traits from language; they are asked to generate language from a target profile. Recent work on virtual personas suggests that extended interview-style backstories can bind models more strongly to target response distributions than thin demographic or biographical prompts (Kang et al., 2025), while work on cross-scale psychometric prediction suggests that models often compress quantitative trait inputs into natural-language summaries before reasoning from them (Liu et al., 2026). Together, these lines point to a central possibility: language may be more than an output surface. It may be a medium through which models organise, carry, and recover individual-difference structure.

Existing evaluations often leave that possibility under-tested. A model can endorse questionnaire items, repeat trait adjectives, or follow a style cue without preserving a coherent profile through open-ended text. We therefore introduce a round-trip evaluation of narrative-mediated psychometric binding. The input is a measured profile from a real participant in PARSEL (Hrkalovic et al., 2025); the output is generated Life Story Interview-style text, using the LSI format as a structured way to elicit long-form first-person narrative (McAdams & McLean, 2013). Independent scorers then receive only the generated text and infer the source psychometric profile. The experiment asks whether measured individual differences remain attached to generated language across two transformations: scores to controlled profile prose, and profile prose to generated life-story narrative.

The study uses two main profile formats: *Psychometric Narrative Profile*, which renders measured psychometrics as prose without personal facts or explicit score language, and *Psychometric Narrative + Facts*, which adds non-diagnostic biographical anchors to the same profile. We generate 4-part and 24-turn LSIs across four main generators and reverse-score every main condition with five independent scorers. Additional appendix sweeps cover broader generator lanes, less-filtered variants, and a diffusion-language model. The contribution is a behavioural assay for psychometric-linguistic encoding and decoding through generated narrative text, not a claim that synthetic LSIs are ecologically equivalent to human interviews.

2 Related Work

Personality prediction from text. Inferring personality from language has a long history in computational linguistics and psychology. The lexical hypothesis posits that personality-relevant distinctions become encoded in natural language (Saucier & Goldberg, 1996), and computational work has used social media, essays, interviews, and spoken or written narratives to recover personality from human-authored text (Boyd & Pennebaker, 2017; Fast & Funder, 2008; Dai et al., 2022; Oltmanns et al., 2026; Speer et al., 2026; Wright et al., 2026). Reported rates vary with genre and dose: Dai et al. (2022) fine-tune InterviewBERT on interview responses, Oltmanns et al. (2026) use McAdams-style narratives, Wright et al. (2026) use zero-shot LLM scoring of stream-of-thought writing, and Speer et al. (2026) compare NLP methods on open-ended responses. These studies establish the recognition side of the problem: personality can be inferred from open human text, though rates depend on genre, length, prompt structure, and scoring method. Our setting reverses the direction. We start from measured psychometrics, generate LSI-style text, and ask what survives the transformation.

LLM persona conditioning. Recent work asks whether language models can adopt stable personas or simulate human response distributions when conditioned on profile information (Argyle et al., 2023; Park et al., 2023; Bai et al., 2025; Serapio-García et al., 2025; Shi et al., 2025; Sorokovikova et al., 2024; Zhou et al., 2025). Most such work uses invented personas, demographic skeletons, thin trait descriptions, or questionnaire self-report by the conditioned model; these choices matter because questionnaire self-report can diverge from task behaviour (Han et al., 2025), and socially grounded persona frameworks require richer information than a trait label alone (Venkit et al., 2026). Kang et al. demonstrate that extended McAdams-style backstories produce deeper persona binding than short descriptions (Kang et al., 2025), while work on cross-scale psychometric prediction finds that models often translate quantitative traits into natural-language summaries before reasoning from them (Liu et al., 2026). These results motivate a stricter generation-and-recovery test of whether measured psychometric profiles bind more strongly when rendered into controlled narrative language.

Life-story interviews and narrative personality. The McAdams Life Story Interview elicits personality through narrative themes rather than direct trait self-description (McAdams, 2001; McAdams & McLean, 2013). Its structure of chapters, key scenes, challenges, values, and future scripts is useful here as a structured prompt format for long-form first-person narrative. Human LSI and self-narration studies indicate that narrative text can carry personality signal (Lodi-Smith et al., 2009; Oltmanns et al., 2026; Wright et al., 2026). At the same time, LLM processing of life narratives can introduce representational shifts and bias (Subbiah et al., 2026). We therefore use the LSI format to structure generated first-person narrative and evaluate recoverable psychometric structure, while reserving matched human-LSI validity for future data collection.

Post-training and mechanism. Work on assistant personas and post-training suggests that chat alignment can privilege a default assistant mode and change behavioural surrogacy (Lu et al., 2026; Binz et al., 2026). Work on model personality and representation also illustrates why mechanism remains difficult: behavioural capability, corpus exposure, and internal representation are related but separable questions (Ju et al., 2025). We therefore pair instruction-tuned systems with base/pretrained controls and OLMo-family corpus inspection, while treating full acquisition provenance as future mechanistic work.

These literatures leave four empirical questions for the present paper:

RQ1. Does measured psychometric structure survive encoding into generated life-story narrative?

RQ2. Is recovery robust to controls against direct score handoff, factual grounding, provider dialect, transparent trait vocabulary, and scorer artefacts?

RQ3. Does recovery appear in base/pretrained generators, or is it created by instruction tuning?

RQ4. Do human readers and short participant-authored conversations provide bounded evidence that generated stories carry person-profile signal?

3 Method

3.1 Data and psychometric targets

We used PARSEL (Hrkalovic et al., 2025), a multimodal dataset collected to study social perception and partner selection in cooperative settings. PARSEL includes 297 participants with psychometric questionnaires, pairwise conversations, and collaborative tasks. We used the $N = 290$ participants with complete profiles for the round-trip analyses. A subset of 248 participants had sufficient conversation data for the human-language reference analyses.

Each participant completed the HEXACO-60 inventory (Ashton & Lee, 2009), yielding Honesty-Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, and Openness. We also used ten non-HEXACO continuous targets: four trust subscales (Propensity, Ability, Integrity, Benevolence; Mayer & Davis 1999), four Psychopathic Personality Traits Scale subscales (Cognitive Responsiveness, Affective Responsiveness, Egocentricity, Interpersonal Manipulation; Boduszek et al. 2016), the Social Interaction Anxiety Scale (Mattick & Clarke, 1998), and social value orientation. We report recovery over three aggregates: HEXACO6, Beyond10, and Continuous16.

3.2 Round-trip pipeline

Profile construction. The first transformation encoded measured scores into natural-language profile text. *Psychometric Narrative Profile* was the clean psychometric condition: it rendered measured scores as prose while excluding personal facts and forbidding explicit instrument names, exact scores, percentiles, construct identifiers, and scoring language where possible. *Psychometric Narrative + Facts* added non-diagnostic biographical anchors to the same psychometric narrative profile. *Facts-Only Control* contained biographical facts without psychometric interpretation and was used as a control rather than as a main treatment condition. Appendix C reports the full profile-writer bake-off and rationale for selecting Gemma 4 31B IT as the default writer.

The profile-writing stage was audited directly rather than treated as hidden prompting infrastructure. Gemma 4 31B IT was selected as the default profile writer after a full-N profile-writer comparison because it preserved strong direct decodability while remaining locally hostable. Appendix C reports the full direct-scoring table for this first transformation.

LSI generation. The second transformation expanded the profile into first-person responses using an adapted LSI prompt format (McAdams & McLean, 2013). The main panel crossed four generators (Gemini 3 Flash, GPT-4.1, Gemma 4 31B IT, and Qwen 3.6 27B), two profile conditions (*PsychNarr* and *PsychNarr+Facts*), and two generation lengths (4-part and 24-turn LSIs). Four-part LSIs tested whether compact narratives already carry recoverable signal; 24-turn LSIs tested the richer long-form setting most aligned with the LSI interview format.

All main rows used $N = 290$. Appendix D reports additional 4-part generator sweeps, including smaller, open-weight, less-filtered, and diffusion-language model lanes.

Blind psychometric scoring. Following recent zero-shot LLM scoring work on personality from open text (Wright et al., 2026), the decoding arm received only the generated LSI text and inferred the source psychometrics. The main scoring panel used Gemini 3 Flash, Sonnet 4.6, GPT-5.4-Mini, Qwen 3.6 Plus, and Gemma 4 31B IT. Scores were compared with the source profiles using Pearson correlations and then averaged over HEXACO6, Beyond10, and Continuous16 targets. Appendix E reports scoring prompts, parsing checks, human-reader calibration, and the PARSEL conversation coding structure.

3.3 Validation and reference analyses

We used several control families to ask what carried the recoverable signal. Direct scorecard controls tested whether exact numeric handoff or construct glossaries recovered the same signal as narrative profile text. Facts-only controls tested whether biographical anchors alone carried psychometric geometry. Family-bias checks compared same-family generator/scorer pairs against cross-family pairs. Additional generator sweeps asked whether the result was confined to a narrow autoregressive chat-model ecology. Transparent schema tests asked how much recovery could be explained by obvious trait-language lexicons. Severe sentence-drop ablations deleted every generated sentence containing any transparent-schema lexical hit before rescoring. Human-reader calibration asked whether non-LLM readers could match short person-description snippets to generated LSI theme snippets above chance.

We also included two reference analyses. True base/pretrained generators tested whether recoverable psychometric-language structure appeared before instruction tuning. PARSEL conversations provided the only participant-authored language tied to the same source profiles. These conversations are short task interactions, so they offered human-language context rather than format-matched validation.

4 Results

We report results in the order of the four research questions. First, we test whether measured psychometric structure survives the full round trip from source profile to generated LSI text and back to inferred scores. Second, we test alternative explanations: direct score handoff, biographical facts, provider-specific generator/scorer effects, and transparent trait vocabulary. Third, we compare true base/pretrained generators with matched post-trained or instruct counterparts to ask whether recovery depends on instruction tuning. Fourth, we report two bounded human-language references: a non-LLM reader matching task on generated snippets and short PARSEL task conversations from the same source participants.

4.1 RQ1: Round-trip recovery across generated LSIs

The first question was whether measured psychometric structure survives the two-step transformation from scores to profile prose to generated life-story narrative. We first scored the profile text directly to estimate how much source structure was available before LSI generation. We then scored the generated LSIs to test how much of that structure survived narrative expansion. The profile-writing transformation was highly informative on its own: Gemini 3 Flash direct scoring of the profile text reached HEXACO/Beyond10/Continuous16 values of .915/.780/.834 for *PsychNarr* and .911/.780/.833 for *PsychNarr+Facts*. The generated-LSI matrix tested the second transformation: whether independent scorers could recover the source psychometrics from generated life-story text alone. It crossed 4 generators, 2 profile conditions, 2 LSI lengths, and 5 scorers, for 80 full-N rows.

Generated LSIs preserved substantial recoverable signal (Table 1; Figure 1). For 24-turn LSIs, *PsychNarr* reached mean HEXACO6/Beyond10/Continuous16 recovery of .787/.619/.682; *PsychNarr+Facts* reached .761/.613/.669. The 4-part LSIs were also strong, with Continuous16 recovery of .659 for *PsychNarr* and .649 for *PsychNarr+Facts*. Longer LSIs improved recovery modestly, but the effect was already visible in compact generated text.

The recovery pattern was not confined to one scorer or generator. Mean Continuous16 recovery across all main rows was .681 for Sonnet 4.6, .672 for Gemini 3 Flash, .666 for Qwen 3.6 Plus, .661 for Gemma 4 31B IT, and .642 for GPT-5.4-Mini. Gemini 3 Flash and Gemma 4 31B IT were strongest as generators, GPT-4.1 remained close, and Qwen 3.6 27B was weaker but still well above null.

Table 1: Round-trip recovery averaged across four LSI generators and five independent scorers.

Condition	LSI length	Rows	HEXACO6	Beyond10	Continuous16
PsychNarr	4-part	20	.755	.601	.659
PsychNarr+Facts	4-part	20	.731	.599	.649
PsychNarr	24-turn	20	.787	.619	.682
PsychNarr+Facts	24-turn	20	.761	.613	.669

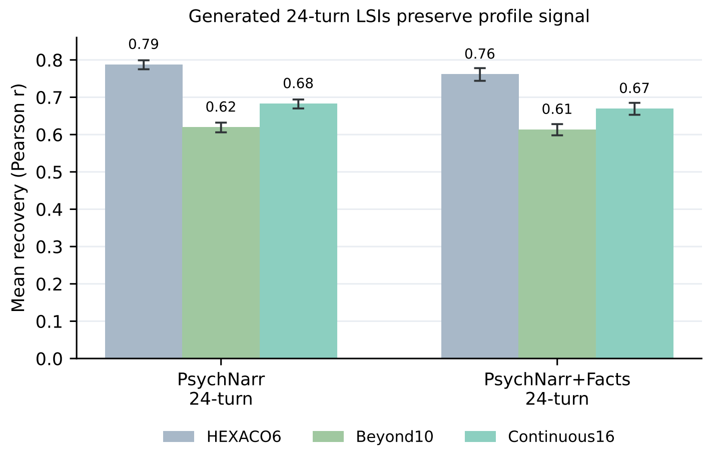

Figure 1: Generated 24-turn LSIs preserve recoverable psychometric signal across both profile formats. Bars show mean Pearson r across generators and scorers; error bars are descriptive intervals over generator-scorer rows.

4.2 RQ2: Controls against handoff, facts, dialect, and transparent schema

The second question was whether recovery could be explained by simpler routes. Direct numeric handoff carried signal, as it should: a compact exact-score scorecard recovered HEXACO/Beyond10/Continuous16 at .654/.515/.567, and a longer glossary scorecard recovered .728/.584/.638 (Table 2). Both scorecard controls recovered less signal than the matched 4-part *PsychNarr* reference using the same two generators and two scorers, especially for Continuous16 and SVO.

Facts-only controls gave the complementary result. Generated facts-only inputs recovered little HEXACO signal: .102 for Gemini-generated facts-only controls and .149 for Gemma-generated facts-only controls. Facts-only rows therefore ruled out biographical anchors as the primary source of recovery in *PsychNarr+Facts*. Same-family generator/scorer cells also did not outperform cross-family cells (HEXACO/Beyond10 .751/.600 versus .760/.610), so the effect was not higher when the same model family generated and scored the text.

Table 2: Controls separating narrative profile conditioning from direct score handoff and facts-only inputs.

Control	HEXACO6	Beyond10	Continuous16	SVO
Matched 4-part *PsychNarr* subset	.761	.618	.672	.572
Compact exact-score scorecard	.654	.515	.567	.412
Glossary scorecard	.728	.584	.638	.549
Gemini facts-only control	.102	–	–	–
Gemma facts-only control	.149	–	–	–

Transparent trait-language controls recovered part of the HEXACO signal. A lexical schema built from public trait and facet descriptors, not model outputs, recovered mean HEXACO6 signal of .511 without an LLM judge. After residualising LLM recovery on all six HEXACO lexicons, mean HEXACO6 recovery remained .668; after residualising on every schema lexicon plus length, it remained .641, with all 192/192 tests significant. A more severe deletion test removed every generated sentence containing any transparent-schema lexical hit before rescoring. For full-N Gemini 3 Flash 24-turn LSIs, this retained only 37.0% of words for *PsychNarr* and 39.9% for *PsychNarr+Facts*, yet HEXACO6 recovery remained .752 and .727 respectively. OLMo sentence-drop controls followed the same pattern. Appendix H reports the lexicon construction, residualisation, sentence-drop tables, and schema-control figure.

4.3 RQ3: Base/pretrained generators and post-training

RQ3 asked whether recovery depends on instruction tuning. To test this, we compared four true base/pretrained generators with matched post-trained or instruct counterparts where a close sibling model was available.

Four true base/pretrained generators recovered HEXACO signal at full $N = 290$: Mistral Small 24B, Gemma 4 31B, Qwen3.5-35B-A3B, and OLMo 3 32B (Figure 2; Appendix G). The Qwen base/instruct comparison uses a different, larger Qwen3.5-35B-A3B family than the Qwen 3.6 27B main generator reported above. Matched post-trained or instruct 4-part LSI runs were higher in all four families under the available scorer coverage: Mistral rose from .453–.461 to .710, Gemma from .467–.522 to .745–.778, Qwen from .645–.672 to .811–.832, and OLMo from .494–.529 to .652.

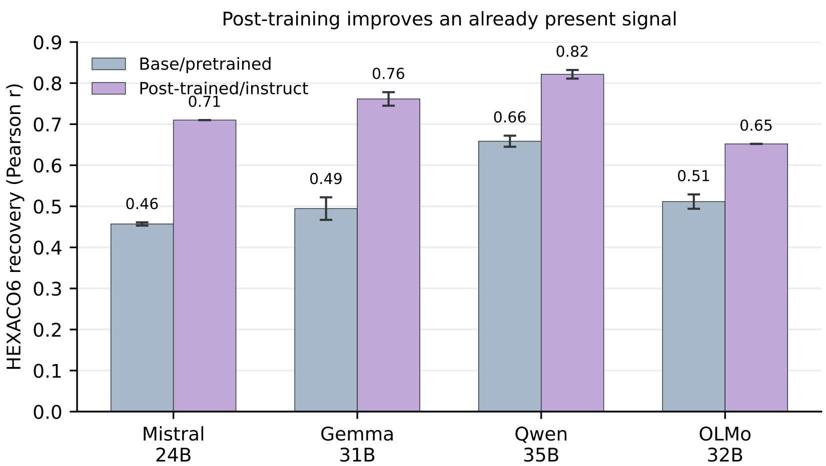

Figure 2: Base/pretrained generators recover HEXACO signal, while matched post-trained or instruct counterparts recover more. Bars show family-level HEXACO recovery; error bars show the available independent-scorer ranges.

4.4 RQ4: Human-reader matching and short-conversation references

The fourth question was whether any evidence outside LLM reverse scoring supports the bounded interpretation that generated stories carry person-profile signal. We first ran a small non-LLM reader calibration. Twenty-five native-English Prolific raters each completed 25 four-option matches between short person-description snippets and generated LSI theme snippets. They selected the correct match on 372/625 trials, for mean accuracy of 59.5% against a 25% chance rate ($t(24) = 8.24$, one-tailed $p = 9.378 \times 10^{-9}$). The task was deliberately lightweight, but it indicates that non-LLM readers can recover person-profile signal from limited generated materials. Appendix E reports the human-reader design and the human-language coding structure used for PARSEL conversations.

The PARSEL conversation references were modest. Participants' own HEXACO scores predicted coded features of their short task conversations at mean $|r| = .094$, with a featurewise best-trait reference of .181. Partner-perceived HEXACO after conversation correlated with self-report at mean $r = .157$, and LLM scoring of the short conversations recovered HEXACO at $r = .179$–$.221$. Against these low-bandwidth references, generated-story-to-real-conversation same-feature correspondence was $r = .147$–$.180$. These values provide the denominator for interpreting the generated-LSI-to-conversation same-feature correlations.

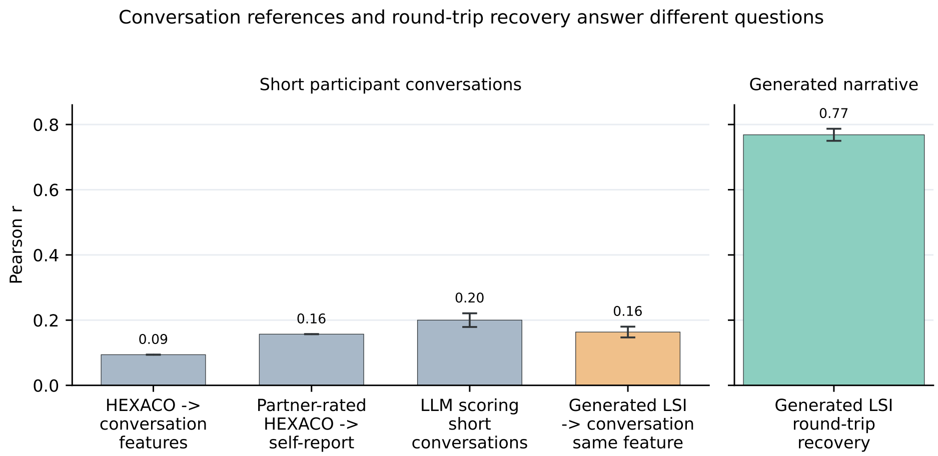

Figure 3: Short PARSEL conversations provide a low-bandwidth participant-language reference. The left panel shows participant-language references derived from short conversations; the right panel shows generated-LSI round-trip recovery, a different construct measured from the full generated narrative.

As a secondary content-level check, profile-conditioned LSIs also had broader emotional-valence variability than unconditioned baselines, and the Sentimentality-to-valence-variability effect replicated across independent valence coders. Appendix F reports the full emotional-reactivity analysis.

5 Discussion

Measured psychometric structure survives transformation into generated life-story narrative. Across the main 24-turn conditions, independent scorers recovered HEXACO traits at $r = .761$–$.787$ and recovered broader individual-difference targets at lower but consistent levels. The simplest shortcuts did not match the full narrative condition: exact scorecards, facts-only controls, same-family generator/scorer pairs, and transparent trait lexicons all recovered less signal than the primary LSI rows. Base/pretrained generators also recovered signal, indicating that instruction tuning is not required, although matched post-trained models performed better. Finally, human readers and short PARSEL conversations provide bounded external references: they support the claim that generated materials carry person-profile signal, while leaving matched human-LSI validation for future work.

The central finding is narrative-mediated psychometric binding in a strictly operational sense: source individual-difference structure remains attached to generated language across transformations. The round trip is therefore both a measurement design and a behavioural assay of two linked capabilities: encoding measured psychometrics into profile and life-story language, and decoding psychometrics back from that language. The effect is robust across multiple generators, multiple scorers, two profile formats, two LSI lengths, direct scorecard controls, facts-only controls, transparent-schema tests, severe sentence deletion, human-reader calibration, true base/pretrained generators, and exploratory generator sweeps that include a diffusion-language model.

The narrative transformation matters because it gives the model an intermediate representation richer than a score table. Compact exact-score scorecards and longer glossary scorecards recover meaningful signal, but they do not match the narrative profile condition. Facts-only controls recover little HEXACO signal. The controlled profile prose can express covariance, tension, and interaction among traits: affective warmth can be separated from anxiety, trust in benevolence can be distinguished from trust in ability, and interpersonal caution can coexist with conscientious self-control. The LSI generator then expands that person-language into episodes, explanations, values, emotional responses, and interpersonal scenes. This interpretation fits adjacent evidence that LLMs can use natural-language summaries as potent psychometric compressions (Liu et al., 2026), while extending it into a stricter generation-and-recovery setting.

The base-model result is especially important for interpreting the capability. The base/pretrained controls indicate that recoverable psychometric-language structure is already available before assistant-style instruction tuning, while matched post-trained/instruct comparisons indicate that post-training can substantially amplify task performance. This pattern complements Kang et al.'s persona-binding result (Kang et al., 2025): deep persona methods are not merely about asking a chat model to role-play, but about activating representational resources that can already be present in pretrained models. It also matters because post-training can reposition models toward a default assistant mode (Lu et al., 2026) and, in some behavioural-surrogate settings, reduce human-response alignment (Binz et al., 2026). Our result makes the profile-to-language mapping a useful probe for future work on how pretraining and post-training shape social-cognitive capabilities.

The schema results should be read constructively. A fine-grained psychometric schema is a plausible mechanism: models may learn rich associations among traits, facets, affective style, interpersonal warmth, anxiety, trust, and language. That possibility does not invalidate the finding. A schema rich enough to predict Sentimentality rather than Anxiety, affective-responsiveness deficits rather than unrelated dark-trait facets, and Benevolence rather than unrelated trust facets is preserving real psychometric structure. The transparent-schema tests indicate that obvious lexical versions of this explanation are incomplete. Richer latent schemas, broader personality-language mappings, and ordinary social regularities in pretraining remain targets for future mechanistic and corpus work.

The participant-language references are deliberately modest. Human open-ended narratives and life narrative interviews can carry recoverable personality signal (Oltmanns et al., 2026; Wright et al., 2026), but those studies use richer and more introspective text than the available PARSEL conversations. Wright et al. report LLM-self-report convergence of $r = .44$–$.48$ from 30-minute stream-of-thought writing and multi-day diaries; the strict PARSEL conversation reference aggregates about 12.7 minutes and 1,110 target-speaker words per participant in short dyadic tasks. We therefore treat the conversation result as a low-bandwidth participant-language reference. It shows that some generated LSI features fall in the same small range as other short-conversation psychometric references, but it does not validate synthetic LSIs as substitutes for human life-story interviews. The secondary emotional-reactivity analysis in Appendix F provides a complementary content-level check that profiles shape narrative dynamics, not only final score recovery.

Kang et al.'s work on deep binding combines behavioural persona tests with corpus checks, not a full provenance account (Kang et al., 2025). Our base-model comparisons and OLMo-family corpus probes belong in that empirical tradition: they narrow and stress-test the interpretation, while leaving mechanistic acquisition as future work. The stronger validation study would collect matched human LSIs, long-form self-narration, or reflective essays from the same participants.

6 Limitations

This study tests psychometric-linguistic encoding and decoding through generated profile prompts and LSI-style text. It does not establish that synthetic LSIs faithfully reproduce how the same people would write or speak in matched human LSIs, and it does not establish unrestricted person simulation. The evidence is English-only and drawn from PARSEL's Western, educated, industrialised, rich, and democratic (WEIRD) participant context, so we do not know whether the same profile-to-language preservation would survive translation, multilingual profile construction and scoring, or lower-resource linguistic contexts. The pipeline is also LLM-dense: language models write profiles, generate LSIs, score psychometrics, and support several analyses. Cross-family scoring, deterministic controls, transparent lexicons, sentence deletion, base models, and human-reader calibration reduce shared-method concerns, but they do not eliminate them.

We also do not identify the training-data source or mechanistic origin of the mapping. OLMo-family corpus tools can inspect ingredients in pretraining data, and base-model controls move part of the phenomenon before instruction tuning, but how language models generalise from training data into social-cognitive capabilities is a much larger open problem. Mapping latent space may reveal the capability's structure; reconstructing exactly how it came to be is beyond this paper. The human-reader pilot is a calibration on generated snippets, not expert annotation of original PARSEL transcripts, and the conversation reference uses short task conversations with modest recoverable psychometric signal. The next validation step is a matched dataset with psychometrics, human LSIs or long-form self-narration, participant-authored conversations, and consent structures designed for both human and synthetic narrative analysis.

7 Conclusion

Stories of Your Life as Others turns narrative-mediated psychometric binding into a measurable behavioural assay. Starting from measured human psychometric profiles, generated LSI-style text preserved recoverable individual differences across a large participant set, multiple model families, independent scorers, profile formats, base-model controls, and schema ablations. The central finding is that language models can transform measured individual-difference profiles into open-ended narrative and that independent models can recover substantial parts of the original psychometric structure from that narrative alone.

That result matters for both social-science simulation and model evaluation. If generated narrative preserves rich profile structure, then LLM personas should be evaluated through behaviourally expressed language, not only questionnaire self-report or short trait labels. At the same time, the controls and human-language references keep the claim bounded: the assay measures profile-to-language preservation, not ecological equivalence to human autobiographical writing. The next step is matched human narrative data that can test how generated LSIs compare with real life-story interviews from the same people.

Reproducibility Statement

All psychometric data are from PARSEL (Hrkalovic et al., 2025). Model versions, prompts, scoring parameters, statistical analysis scripts, and derived results are staged in the accompanying repository. The manuscript reports values from the consolidated artefact package, with table and figure provenance documented in the appendix map and evidence manifests.

Ethics Statement

This study uses existing psychometric data collected under ethical approval. No directly identifiable information appears in the manuscript. *Psychometric Narrative + Facts* profiles include source-person life anchors for experimental conditioning; we therefore treat them as controlled research artefacts and distinguish them from *Psychometric Narrative Profile*, which removes personal facts. The capacity to recover personality information from generated text raises privacy considerations that warrant careful attention in deployment contexts.

AI Disclosure Statement

As the generative pipeline is itself the subject of this research, large language models served as experimental instruments throughout: as profile writers, narrative generators, personality scorers, content coders, matchers, leakage auditors, and corpus-inspection aids. The human authors were additionally aided in literature search, coding support, statistical checking, and manuscript preparation by generative agents operating within customised coding harnesses. Generated narratives were evaluated for psychometric signal recovery and behavioural correspondence, not subjective quality alone. All methodological decisions, experimental design choices, and interpretive claims are by the human authors, who are responsible for the accuracy, academic integrity, and intellectual contributions of this paper.

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A Appendix Roadmap

This appendix is an evidence map for the revised manuscript. The main text reports the compact argument; the appendix records the controls, implementation routes, and interpretation boundaries needed to audit the result. The accompanying repository contains the full prompts, generated outputs, scorer rows, parser outputs, and statistical scripts.

Appendix section	Purpose	Key evidence route
Profile conditions	Defines the public condition names and maps them to internal run labels.	Appendix B.
Profile writer bake-off	Documents that the score-to-profile transformation was selected empirically, not treated as hidden prompt craft.	Appendix C; full-*N* profile-writer rows scored by Gemini 3 Flash.
Model sweep	Documents the central 80-row matrix, broader 4-part generator lanes, less-filtered variants, and diffusion-language-model check.	Appendix D; consolidated run manifests.
Scoring and coding	Documents reverse scoring, parsing checks, human-reader calibration, and PARSEL conversation denominators.	Appendix E.
Emotional reactivity	Reports the secondary content-level analysis of valence variability and independent valence coders.	Appendix F.
Base and post-trained generators	Separates pretraining-available capability from post-training amplification.	Appendix G; full-*N* base/pretrained and matched instruct/post-trained rows.
Schema and sentence deletion	Tests whether obvious trait-language schemas explain recovery.	Appendix H; lexicon construction, residualisation, and sentence-drop outputs.
Corpus inspection	Adapts Kang-style corpus checks and OLMo-Trace/InfiniGram probes to generated LSI outputs and schema-hit sentences.	Appendix I; `corpus_inspection_bundle_20260608/README.md`.

Table 3: Camera-ready appendix map. The table is a routing surface for the larger evidence forest, not a replacement for the linked manifests and analysis outputs.

B Profile Conditions and Terminology

The paper uses public-facing condition names in the main text while preserving older internal run labels in the artefact tree. The public names are deliberately descriptive.

Public condition	Short label	Meaning
Psychometric Narrative Profile	*PsychNarr*	A controlled prose profile constructed from measured psychometrics. It excludes personal facts and avoids explicit instrument names, exact scores, percentiles, and scoring language where possible.
Psychometric Narrative + Facts	*PsychNarr+Facts*	The same psychometric narrative profile, augmented with non-diagnostic biographical anchors. Legacy artefact labels include “FullBio” and “ArtWeave”.
Facts-Only Control	*FactsOnly*	Biographical anchors without psychometric interpretation. This is a control, not a main treatment condition.
Scorecard controls	Scorecard	Deterministic, non-narrative source-data handoffs: compact exact-score rows and a longer glossary variant.

Table 4: Condition names used in the revised paper.

C Profile Writer Bake-Off

The main experiments use Gemma 4 31B IT to write the psychometric narrative profiles. This choice followed a full-*N* comparison of candidate profile writers. The strongest direct profile decode came from GPT-5.5, but its *PsychNarr+Facts* profiles were much longer. Gemma 4 31B IT preserved high direct decodability while producing compact profiles and remaining open-weight and locally hostable.

Profile writer/condition	N	Mean words	Median words	HEXACO	Beyond10	PPTS total
Gemma 4 31B IT *PsychNarr*	290	930	929	.915	.780	.870
Gemma 4 31B IT *PsychNarr+Facts*	290	1115	1114	.911	.780	.874
Opus 4.7 *PsychNarr*	290	1060	1059	.921	.744	.771
Opus 4.7 *PsychNarr+Facts*	290	1395	1389	.900	.721	.768
Qwen 3.6 *PsychNarr*	290	1192	1186	.907	.743	.822
Qwen 3.6 *PsychNarr+Facts*	290	1627	1520	.892	.740	.824
Qwen 3.7 *PsychNarr*	290	1111	1106	.917	.738	.782
Qwen 3.7 *PsychNarr+Facts*	290	1603	1580	.910	.732	.788
GPT-5.5 *PsychNarr*	290	1524	1522	.922	.798	.880
GPT-5.5 *PsychNarr+Facts*	290	2940	2932	.915	.787	.879

Table 5: Full-N profile-writer comparison scored by Gemini 3 Flash. These rows evaluate the first transformation directly: measured psychometrics to controlled profile language.

D Model Sweep and Generation Variants

The main matrix is the 80-row panel summarised in Table 1: four LSI generators, two profile formats, two LSI lengths, and five independent scorers. All rows use $N = 290$. The main generators are Gemini 3 Flash, GPT-4.1, Gemma 4 31B IT, and Qwen 3.6 27B; the main scorers are Gemini 3 Flash, Sonnet 4.6, GPT-5.4-Mini, Qwen 3.6 Plus, and Gemma 4 31B IT.

Additional 4-part sweeps were used to stress-test the ecology around that main panel. These include smaller and open-weight generators, safety-filtered and less-filtered ("Heretic") variants, and Mercury, a diffusion language model. The generator sweep is not used to inflate the headline; it is used to check whether recovery is an artefact of one narrow provider or one autoregressive chat-model style.

Sweep family	Question	Summary
Main 80-row matrix	Does recovery survive model/scorer variation?	Four generators, two profile formats, two LSI lengths, five scorers, all full-N. Same-family generator/scorer cells do not outperform cross-family cells on HEXACO/Beyond10 (.751/.600 vs. .760/.610).
Broader 4-part lanes	Is the result limited to the four main generators?	Additional local and API lanes include smaller, open-weight, and alternative provider models. These are appendix robustness checks rather than headline rows.
Less-filtered variants	Is ordinary safety filtering suppressing or manufacturing recovery?	Full-N *PsychNarr+Facts* 4-part comparison changed HEXACO6 by $-.003$ for Gemma and $+.045$ for Qwen across four judges.
Diffusion-language model	Is recovery specific to autoregressive generation?	The broader sweep includes Mercury as a diffusion-language-model lane; this is treated as exploratory evidence against a narrow autoregressive artefact.

Table 6: Generation-variant checks supporting the main matrix.

E Scoring, Human Reader Calibration, and Conversation Coding

The main reverse-scoring panel uses five independent LLM scorers. Each scorer receives only the generated LSI text and predicts the source psychometric targets. Scoring follows the zero-shot open-text personality-scoring logic of Wright et al. (2026), adapted to generated LSI-style responses and to the HEXACO plus beyond-HEXACO target set. Parsed scorer outputs were checked for row count, participant identifiers, missing targets, and out-of-range values before correlations were computed.

The human-reader calibration asks a narrower question: can non-LLM readers detect person-profile signal in short generated materials? Twenty-five native-English Prolific raters completed 625 four-option matches between short person-description snippets and generated LSI theme snippets. They selected the correct match on 372 trials, for mean accuracy of 59.5% against a 25% chance rate ($t(24) = 8.24$, one-tailed $p = 9.378 \times 10^{-9}$). This is not matched human-LSI validation; it is a calibration that the generated materials carry person-profile signal visible to human readers.

Table 7: Human-reader calibration on short snippets of generated materials.

Item	Value
Native-English Prolific raters	25
Four-option matches completed	625
Correct matches	372
Mean accuracy	59.5%
Chance accuracy	25%
One-sample test	$t(24) = 8.24$, $p = 9.378 \times 10^{-9}$
Attention checks	75/75 passed
Mean completion time	60.6 minutes

The PARSEL conversation analyses use the only participant-authored language tied to the same source profiles. Conversations are short cooperative-task interactions, so the reference is intentionally low-bandwidth. Participants' own HEXACO scores predict coded conversation features at mean $|r| = .094$, with a featurewise best-trait reference of .181. Partner-perceived HEXACO after conversation correlates with self-report at mean $r = .157$, and LLM scoring of the short conversations recovers HEXACO at $r = .179$–$.221$. These denominators keep the generated-LSI-to-conversation bridge in proportion.

Table 8: PARSEL conversation references. Conversations are short task interactions, so they provide low-bandwidth human-language context.

Reference	Pearson r
Source HEXACO $\rightarrow$ coded conversation features	.094
Featurewise best-trait reference	.181
Partner-perceived HEXACO $\rightarrow$ self-report	.157
LLM scoring of short conversations	.179–.221
Generated LSI $\rightarrow$ conversation same-feature bridge	.147–.180

F Generated Emotional Reactivity

The emotional-reactivity analysis tests whether profile conditioning affects generated narrative dynamics, not only final psychometric scores. Profile-conditioned LSIs vary more in emotional-valence variability than unconditioned LSIs: conditioned lanes have between-story SDs of .102–.131, while unconditioned baselines range from .070–.088 (Figure 4). Brown-Forsythe tests are significant in 27/32 conditioned-versus-unconditioned comparisons, while unconditioned baselines do not differ from each other.

The Sentimentality-to-valence-variability effect also replicates across independent valence coders, with mean partial correlations controlling for mean valence of .449 for Qwen 3.7 Max, .450 for Grok 4.20, and .424 for Xiaomi MiMo. These content-level checks are secondary to the main round-trip recovery result, but they indicate that the profile shapes generated narrative dynamics as well as scorer-level trait estimates.

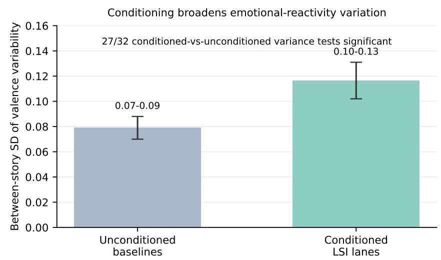

Figure 4: Unconditioned generated narratives have narrower emotional-valence variability than profile-conditioned LSI lanes.

G Base/Pretrained and Post-Trained Generators

The base/pretrained analysis uses a full four-family panel with matched post-trained or instruct 4-part counterparts where available. The central interpretation is qualitative as well as numerical: the capability is not created by instruction-tuned chat compliance, but post-training makes the task stronger and easier to control.

Table 9: Matched base/pretrained and post-trained/instruct 4-part LSI controls. Values are HEXACO6 recovery; ranges reflect available independent scorers and available scorer coverage differs slightly by family.

Model family	Base/pretrained	Post-trained/instruct
Mistral Small 24B	.453–.461	.710
Gemma 4 31B	.467–.522	.745–.778
Qwen3.5-35B-A3B	.645–.672	.811–.832
OLMo 3 32B	.494–.529	.652

The base/pretrained rows use true base or pretrained checkpoints for Mistral Small 24B, Gemma 4 31B, Qwen3.5-35B-A3B, and OLMo 3 32B. The Qwen family in this base/instruct comparison is not the Qwen 3.6 27B checkpoint used as one of the four main LSI generators; it is a different, larger Qwen3.5-35B-A3B sibling pair. The matched post-trained/instruct values use the closest available sibling rows under available scorer coverage. Scorer parity is strongest for the Qwen family; Mistral and OLMo instruct values are Gemini-scored, and Gemma IT values come from the main 4-part *PsychNarr+Facts* scorer matrix.

H Transparent Schema and Sentence-Drop Tests

Transparent lexical schemas were built from public trait and facet descriptors rather than from model outputs. They intentionally operationalise the simplest version of a psychometric-schema account: if obvious public trait language carries the result, a non-LLM lexical scoring scheme should recover much of the same signal. It does recover a real part of the signal. Across eight headline 24-turn lanes, the transparent schema alone recovers mean HEXACO6 signal of .511. But the LLM-scoring result remains substantial after residualising on the schema scores: raw HEXACO6 recovery averages .774, partial recovery after all six HEXACO lexicons is .668, and partial recovery after every schema lexicon plus length is .641, with 192/192 tests significant.

The sentence-drop control is deliberately more severe than literal overlap deletion. Every generated sentence containing any transparent-schema lexical hit is removed before rescoring. For Gemini 3 Flash 24-turn LSIs, this leaves no transparent schema hits and retains only 37.0% of words for *PsychNarr* and 39.9% for *PsychNarr+Facts*. Recovery survives. For OLMo base LSIs, deletion retains 56.1% of words; for OLMo instruct LSIs, deletion retains 43.4% of words. These controls do not rule out a rich latent schema. They indicate that the obvious lexical version is incomplete.

Table 10: Transparent-schema controls. Sentence-drop rows delete every sentence containing any transparent-schema lexical hit before rescoring.

Test	HEXACO6	Note
Raw LLM scoring, 8 main 24-turn lanes	.774	Original recovery
Transparent lexical schema alone	.511	No LLM judge
Partial after all six HEXACO lexicons	.668	Residualised
Partial after all schema lexicons + length	.641	Residualised, 192/192 significant
Gemini *PsychNarr* sentence drop	.813 → .752	37.0% words retained
Gemini *PsychNarr+Facts* sentence drop	.794 → .727	39.9% words retained
OLMo base sentence drop	.529 → .453	56.1% words retained
OLMo instruct sentence drop	.652 → .567	43.4% words retained

We also ran non-LLM lexical-valence checks using Empath-style lexical categories on base-model outputs. These analyses are secondary because Empath is blunt relative to the LLM scorers, but they help separate the content result from LLM-only scoring. The staged outputs live under `corpus_inspection_bundle_20260608/14_base_empath_variability/`.

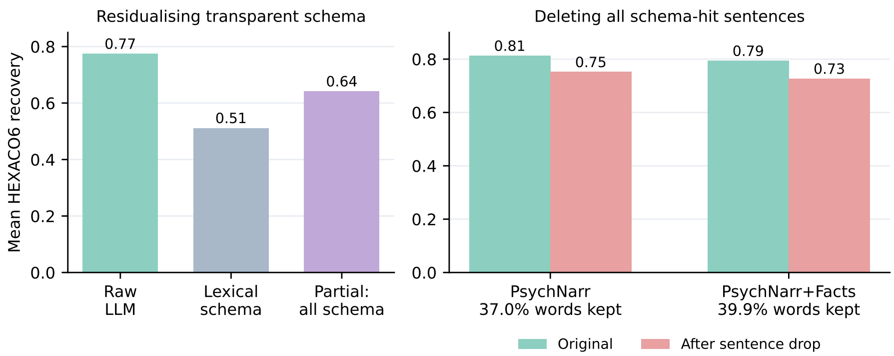

Figure 5: Transparent-schema tests. Left: obvious lexical schemas recover part of the signal, but substantial recovery remains after residualising on schema lexicons. Right: recovery remains after deleting every sentence containing any transparent-schema lexical hit.

I Corpus Inspection Bundle

The OLMo inspection work lives in `corpus_inspection_bundle_20260608/`. The bundle uses symlinks to preserve provenance to the original analysis outputs while providing a compact retrieval surface. Its front door is `corpus_inspection_bundle_20260608/README.md`; its itemised map is `corpus_inspection_bundle_20260608/MANIFEST.tsv`. The purpose is bounded corpus inspection, not provenance proof.

I.1 Kang-style generated-output n-grams

Following the spirit of Kang et al.'s generated-output corpus appendix, we computed generated 5-grams and 10-grams from four generated-LSI corpora and queried exact counts against two OLMo-family InfiniGram indexes. No raw PARSEL conversations or source profile prompts were submitted to InfiniGram; queries use generated LSI surface strings only.

Generated corpus	Files	Tokens	10-grams	Notes
Gemini 24-turn *PsychNarr*	290	2,934,047	2,931,437	Main psychometric narrative profile lane.
Gemini 24-turn *PsychNarr+Facts*	290	2,836,886	2,834,276	Main psychometric narrative plus facts lane.
OLMo base 4-part	290	836,708	834,098	Full-*N* OLMo base lane used for closed-loop probes.
OLMo instruct 4-part	290	414,906	412,296	Full-*N* OLMo instruct lane assembled from 40-PID and remaining-250 runs.

Table 11: Generated corpora used for Kang-style n-gram inspection.

Short 5-grams often appear in OLMo-family indexes because they are ordinary narrative phrases. Generated 10-grams are much less often exact pretraining strings. Among the top 20 generated 10-gram rows per corpus, zero exact hits were found in both checked OLMo-family indexes for 18/20 *PsychNarr* rows, 19/20 *PsychNarr+Facts* rows, 12/20 OLMo base rows, and 20/20 OLMo instruct rows. Anchor-conditioned 10-grams had the same pattern: 32/33, 32/33, 26/33, and 31/33 rows respectively had zero exact hits in both checked indexes. The full tables are in `analysis/kang_style_corpus_appendix_20260608/`.

I.2 Psychometric bridge phrase counts

Exact-count bridge probes found individual ingredients in the checked OLMo-family indexes, but not obvious recipe phrases linking the instruments and LSI format.

Exact phrase	Count
`HEXACO`	22,615
`Life Story Interview`	453
`personal narrative`	509,630
`Psychopathic Personality Traits Scale`	135
`Social Interaction Anxiety Scale`	1,335
`Social Value Orientation`	1,584
`HEXACO Life Story Interview`	0
`HEXACO personal narrative`	0
`HEXACO Psychopathic Personality Traits Scale`	0
`HEXACO Trust benevolence`	0

Table 12: Representative exact-count probes over checked OLMo-family indexes. Counts support an "ingredients without obvious recipe" interpretation, not an absence-of-evidence claim over all possible training text.

Covariance-literature probes had a similar split. In the checked exact phrase set, explicit bridge phrases produced 0/30 nonzero exact phrases (total count 0), ingredient phrases produced 22/32 nonzero exact phrases (total count 32,410), and literature-handle phrases produced 0/10 nonzero exact phrases. The full query set is staged under `corpus_inspection_bundle_20260608/04_covariance_literature_probe/`.

I.3 OLMoTrace closed-loop checks

We then traced the sentences most favourable to the transparent-schema concern: deleted sentences from OLMo-family outputs that contained transparent schema hits. For OLMo base, 40/40 endpoint calls succeeded and returned 1,550 source rows. Returned sources were diffuse: uncategorised web text (31.9%), education/jobs (20.9%), health (16.1%), literature (10.1%), finance/business (4.8%), art/design (4.2%), politics (3.5%), and science/technology (3.4%). Returned matched spans did not contain explicit psychometric bridge language. For OLMo instruct, 40/40 endpoint calls succeeded and returned 1,547 source rows; a strict scan found zero explicit psychometric labels or bridge handles in returned span text or source URIs.

These traces are deliberately modest. They indicate that the trait-bearing deleted sentences most favourable to a transparent-schema account trace to broad ordinary web, life, affect, health, education, literary, and topical sources rather than obvious psychometric bridge recipes. They do not identify how OLMo learned the capability, rule out all relevant corpus examples, or locate internal features.

I.4 Corpus-inspection boundary

The corpus-inspection bundle can support four bounded conclusions. First, generated LSI text uses ordinary narrative ingredients. Second, common generated longer strings are generally not verbatim pretraining copies in the checked OLMo-family indexes. Third, obvious exact psychometric-to-LSI bridge phrases and covariance recipe phrases were not found in the checked exact-count probes, even though individual ingredients were present.

Fourth, the OLMo-family sentence-drop controls indicate that recovery survives deletion of the most transparent trait-language sentences at full N.

It cannot establish the exact training-data source of the psychometric-language mapping, prove absence of all relevant examples, or make mechanistic claims about circuits, features, attention heads, or latent activation geometry. Those are future work. The present paper uses corpus inspection the way the main method uses behavioural controls: to bound simpler explanations while keeping the acquisition question open.

J Evidence Manifest

The consolidated evidence forest is larger than the manuscript can reproduce. The following routes are the intended audit surfaces for revision and follow-on cleanup.

Route	Contents
`00_front_door/LOAD_BEARING_INDEX.md`	Front-door index for the rebuttal evidence package, including main runs, controls, and analysis outputs.
`00_front_door/FINAL_TABLE_STAGING_20260525.md`	Older table staging surface used to reconcile submitted-paper claims with CCV1 results.
`00_front_door/NUMBER_TO_ARTIFACT_MANIFEST_20260525.md`	Number-to-artifact route map for rebuttal figures, tables, and checks.
`tables/` and `figures/camera_ready/`	Camera-ready table snippets and rebuilt main figures.
`analysis/kang_style_corpus_appendix_20260608/`	Generated n-gram and InfiniGram exact-count outputs for corpus inspection.
`corpus_inspection_bundle_20260608/`	Compact public-facing bundle for OLMoTrace/InfiniGram, schema, sentence-drop, base-control, and covariance probes.

Table 13: Primary audit routes for the revised evidence package.